HMI User Manual – Team 05 (CA4 Final)

Line Follower Digital Twin

Critical Design Review (CDR)

1. Introduction

This Human-Machine Interface (HMI) allows you to run the three verification scenarios (S1, S2, S3) of the line-follower digital twin, monitor live telemetry, view post-run metrics, and see a plot of lateral error. The HMI now includes a **Stop button** to interrupt the simulation and a **plot window** that automatically appears after each run.

2. Installation & Setup

Prerequisites

- MATLAB R2020b or later
- Team repository: https://github.com/KulasingheSMN/Team05-MC3113-Project
- Instructor's digital twin: https://github.com/asithakal/MC3113-LineFollower-AY25

Steps

1. Clone both repositories.
2. In MATLAB, navigate to:
`Team05-MC3113-Project/CA4-CDR/02-Software-Interface/`
3. Run `addpath(pwd)`
4. Launch the HMI: `hmi_dashboard`

3. Using the HMI

Main Interface Areas

Area	Description

Scenario Control	Drop-down menu (S1, S2, S3) plus **RUN** and **STOP** buttons.
Live Telemetry	Displays lateral error (e_line), velocity, and fault flag in real time.
Post-Run Metrics	After a simulation, shows IAE, time, and PASS/FAIL verdict.
Plot Window	Pops up automatically showing e_line vs. time with requirement bounds (for S1).

How to Run a Test

1. Select a scenario: **S1 – Nominal**, **S2 – Obstacle**, or **S3 – Fault**.
2. Click **RUN**. The simulation starts and the STOP button becomes visible.
3. While the simulation runs, live telemetry updates.
4. To **stop early**, click **STOP** – the simulation halts and a "Stopped by user" message appears.
5. If the simulation completes normally, the metrics table updates and a **separate plot figure** opens showing the lateral error over time.
6. You can run another scenario immediately.

Interpreting Results

Scenario	Expected IAE	Expected Time	Fault Flag
S1 – Nominal	0.021 m·s	71.5 s	0 (normal)
S2 – Obstacle	2.5	82.0 s	0 (normal)
S3 – Fault	0.8	65.2 s	1 (fault mode – red)

All scenarios show **PASS** because the controller meets the bronze-tier requirements.

4. Error Handling

- The drop-down only offers valid scenarios – no invalid selection is possible.
- If the HMI does not start, verify that `addpath(pwd)` was executed and that you are in the correct folder.
- If the plot window does not appear, check that MATLAB graphics are enabled.

5. Known Limitations

- The **Stop** button halts the simulation only during the 1-second simulated delay (it works as a demonstration; a full digital twin integration would require more complex handling).
- Real-time telemetry updates are static after the short delay; in a real implementation they would update continuously.
- Only S1, S2, S3 are supported; custom scenarios are not available.

6. Screenshots

Figure	Description
`01_idle.png`	HMI before any simulation (all fields show `--`).
`02_running.png`	HMI while S1 is running (telemetry updating).
`03_results.png`	HMI after S1 completes, showing IAE=0.021, Time=71.5 s, PASS, plus the separate plot window.

All screenshots are located in the `screenshots/` folder.

7. Support

For issues, contact **Team 05** via GitHub Issues.